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Research Agenda

My research produces scholarship and regulatory proposals at the intersection of law and technology through two related strands. First, I examine how artificial intelligence (AI) systems affect established legal frameworks such as intellectual property, administrative law, tort, and privacy. Second, I use computational and empirical methods to study legal change and evaluate emerging technologies and their institutional consequences. This work requires integrating technical expertise with legal scholarship to address questions that transcend traditional disciplinary boundaries.

This interdisciplinary approach draws on my extensive technical expertise in software development, spanning nearly a decade as a senior software engineer in Big Tech (e.g., at Cisco and Lyft), where I developed patented machine-learning features and deployed them on distributed computational clusters, reaching millions of users across markets worldwide. Such a technical foundation complements my experience in patent litigation, federal district and appellate courts, and the United States Senate, enabling me to identify and address a critical gap in contemporary legal scholarship—the need for interdisciplinary research that not only examines how law should adapt to technological changes, but also translates complex technical systems into clear, accessible concepts for legal and policy audiences, as well as students navigating these emerging issues. Without such research, AI governance risks remaining mired in technical opacity, failing to keep pace with technological developments and excluding meaningful democratic participation.

To that end, I propose a multi-tiered approach to studying and governing AI that both increases transparency and keeps pace with the rapid development of generative AI. This approach differs from existing frameworks that focus primarily on model design or algorithmic outputs from systems such as ChatGPT, Claude, and Gemini. Instead, it looks beyond the model to the broader technical and institutional ecosystem in which AI systems are developed, evaluated, and deployed—including software-development processes, software scaffolding, computational infrastructure providers, benchmarking practices, user interfaces, and institutional decisionmaking procedures.¹ Focusing on this multifaceted ecosystem can improve our understanding of AI systems by shifting the legal gaze from opaque models to the more transparent and governable actors, processes, and infrastructures surrounding them. It also reveals how AI can alter established legal doctrines and institutional practices even when the model itself is not the direct object of regulation. More importantly, this approach identifies intervention points that are less subject to the rapid pace of model development, thereby supporting more durable and scalable legal responses across different technologies, institutions, and deployment contexts.

I. AI, Legal Institutions, and Regulatory Governance

The first strand of my research examines how AI and related technical systems relocate consequential choices across software development, technical infrastructure, and institutional decisionmaking. My published work traces this problem through bias in software practices, rule-based code, and infrastructure providers. My current projects extend that framework to administrative decisionmaking and the patent system, asking how legal safeguards should respond when technical processes structure the evidence, technical capabilities, and disclosures on which legal decisions and rights depend.

A. Algorithmic Bias in the Software Development Process

In *Governing the Algorithmic Turn: Lyft, Uber, and Disparate Impact*, Wm. & Mary Ctr. L. & Ct. Tech. (2022), I analyzed how biases embedded in the Agile software development process at transportation

¹ I use “software scaffolding” to refer to the non-model software surrounding an AI model that supplies data and context, processes model outputs, connects the model to tools and external systems, and governs how those outputs are translated into actions. Additionally, I use “computational infrastructure providers” to refer to companies that provide the computational power to train AI models.

network companies like Lyft and Uber can produce discriminatory outcomes. For example, technical decisions in these systems can unintentionally result in higher prices in low-income neighborhoods. My analysis revealed that these discriminatory outcomes stem from both technical decisions (e.g., geohashing user data) and non-technical choices (e.g., region-specific algorithmic tuning), which introduce human and systemic biases into the software development process. I proposed dual remedial paths: immediate relief through disparate impact litigation under Title II of the Civil Rights Act and the Americans with Disabilities Act, and long-term solutions through algorithmic standards of care developed by technical consortia like the Institute of Electrical and Electronics Engineers, which would also enhance algorithmic transparency.

B. Regulation of Rule-based Algorithms

In *Engineering a Fiduciary: Expanding the Regulatory Scope of Algorithmic Bias*, Harv. J.L. & Tech. Dig. (2024), I identified critical gaps in existing and proposed regulatory frameworks that focus predominantly on machine learning (ML) algorithms while neglecting biases inherent in non-ML software code. Through an analysis of Google Chrome's Accessibility feature on Android, I demonstrated how demographic homogeneity within engineering teams can introduce implicit bias that embeds and perpetuates structural inequalities. To illustrate, I analyzed an internal Google proposal to remove an accessibility feature used by blind users, a recommendation predicated on the unfounded assumption that such functionality was unnecessary. To address these challenges, I advocated imposing a fiduciary duty on software engineers, contending that professional self-regulation not only defrays the cost of external regulation but also preserves the profession's autonomy, thereby guarding against potential governmental overreach.

C. Generative AI and Infrastructure Governance

My work on generative AI extends this systems-oriented approach beyond software-development practices and rule-based code to the infrastructure and other technical components that enable AI systems to be developed and deployed. Rather than treating the model or its outputs as the sole objects of legal scrutiny, I examine where control, observability, and the capacity to mitigate harm reside across the broader AI ecosystem. Infrastructure providers are especially significant because they supply the computational resources necessary to train and operate generative AI models and may be better positioned than individual users to implement safeguards, preserve information, and reduce misuse.

In *Audio Deepfakes and the Regulation of the Landlords of Creativity* (with George He), Cambridge F. AI: L. & Governance (2025), I conducted a comprehensive analysis of how regulatory regimes such as the American Blueprint for an AI Bill of Rights, the EU AI Act, and the Chinese Algorithm Recommendation for Internet Information Services address misuses of generative AI. I proposed holding the companies with the hardware capabilities to train generative AI models ("landlords of creativity") liable for misuses of said models because they are the cheapest cost avoiders—they not only own all the critical infrastructure to train the models, but also own the models themselves. I recommended that the landlords could reduce liability if they (1) limit the features provided to unidentified customers and/or (2) maintain an audit trail of generated outputs for registered users, including the attribution and identification of user inputs for generated content.

This article established the infrastructure-focused component of my broader research agenda. My current and planned work extends that inquiry beyond infrastructure providers to the other technical and institutional components surrounding AI models. These projects examine how software scaffolding enables agentic conduct, how AI-assisted processes structure agency decisionmaking, and how the technical production of model capabilities complicates patent disclosure. Together, they ask how law can locate responsibility and preserve accountability when consequential choices are distributed across models, software systems, infrastructure, and institutions.

D. Current and Future Research

Shadow Judgment: AI and the Administrative Record

My current article, *Shadow Judgment: AI and the Administrative Record*, examines agency use of AI systems that rank evidence, classify matters, or recommend action while final legal authority remains with a human official. Although the official formally makes the decision, these systems can shape what evidence the official sees, how that evidence is ranked, and which options appear salient, even when the systems' role is not fully reflected in the administrative record. I call this upstream technical influence "shadow judgment."

Using AI-assisted patent examination as a mechanism-tracing case study, the article audits prosecution histories to compare AI-generated search results, formal citations, and the agency's stated reasons. Drawing on those examples, the article develops two complementary legal doctrines. "Reasoned reliance" requires an agency to explain when and why it relied on an AI-generated premise, with more explanation required when that premise was important to the outcome, difficult to verify, or especially prone to error. Genuine independent review by a human decisionmaker can reduce that burden. "Functional incorporation" requires the agency to include enough information about an AI-assisted process in the administrative record when that information is necessary for a court to evaluate a properly preserved claim of error or prejudice. Together, these doctrines extend administrative-law safeguards to consequential technical processes without requiring agencies to preserve every artifact their systems produce.

Generative AI and the Limits of Patent Law

A second current project, *Generative AI and the Limits of Patent Law*, examines whether Patent Act § 112 can sustain the patent bargain when valuable AI inventions reside not in a stable artifact but in a process that uses data, training methods, evaluation, and infrastructure to produce models with measurable capabilities. Conventional enablement doctrine asks whether a disclosure permits a skilled reader to make and use the claimed invention. Generative AI complicates that inquiry because identical instructions may not reproduce identical model weights or outputs, while disclosure of raw data and proprietary weights may be unnecessary to communicate the inventive contribution.

I develop a process- and performance-based account of enablement that would require disclosure of a reproducible training methodology and measurable performance characteristics rather than a frozen model or every proprietary input. The project also examines the Defend Trade Secrets Act as the alternative legal regime governing model weights, datasets, training pipelines, evaluation methods, and deployment know-how. Its empirical extension will test whether the mismatch between generative AI and patent disclosure is shifting commercially valuable technical knowledge from patents into trade secrecy, contracts, and infrastructure control. By connecting § 112 doctrine to firms' protection choices, the project evaluates whether current innovation policy is preserving the disclosure bargain or unintentionally accelerating an AI secrecy shift.

Governing Agentic AI Through Software Scaffolding

Agentic AI systems sharpen the need to look beyond the model itself. A foundation model does not independently browse the internet, execute code, access private databases, make purchases, or take actions within an organization. Those capabilities arise from the software scaffolding that connects the model to tools, memory, data, and external systems. That scaffolding also determines the system's permissions, action limits, approval requirements, monitoring mechanisms, and audit trails. I will examine whether regulating these surrounding components offers a more durable and technically tractable means of addressing the risks associated with increasingly autonomous AI systems. Rather than attempting to regulate every model capability or unpredictable output, policymakers could impose obligations at the points where model-generated recommendations are converted into real-world actions—for example, by requiring permission controls, human authorization for consequential acts, persistent logging, rollback mechanisms, and limits on tool and data access. This project extends my earlier work on non-ML code and infrastructure providers by

showing how software scaffolding transforms a probabilistic model into an operational actor and therefore constitutes a critical site of legal responsibility.

II. Empirical Legal Studies

I use computational text analysis, machine learning, and natural language processing to study law and legal institutions and to test claims that doctrinal analysis alone cannot resolve. Drawing on my legal training and interdisciplinary background in science, technology, and society, I collaborate with scholars in computer science, economics, and the humanities on projects that evaluate emerging technologies, legal change, and the reliability of the tools increasingly used to study and practice law. Some of this work uses computational methods to measure legal phenomena that conventional proxies overlook. Other projects turn the empirical lens toward the tools themselves, examining what they measure, when their results are reliable, and how methodological choices shape conclusions about their capabilities.

A. Integrating Computational Text Analysis into Legal Research

I co-authored two articles with Professor Michael A. Livermore (UVA Law)—*Studying Judicial Behavior with Text Analysis*, Oxford Handbook Compar. J. Behav. (2023) and *Computational Legal Studies Comes of Age*, Eur. J. Empirical Legal Stud. (2024)—surveying how scholars used various computational text analysis techniques to study legal phenomena. While recognizing the transformative potential of these digital tools, we also identified critical limitations that scholars must address to ensure reliable results. As an example, we pointed out that many such programs are primarily designed by English speakers to analyze English texts, making them poorly suited for analyzing non-English documents. Similarly, we cautioned against using tools designed for one legal domain (e.g., transactional law) to investigate phenomena in another (e.g., litigation). Most importantly, we advised researchers to double-check their inputs and outputs to avoid the problem of garbage in, garbage out.

B. A Case Study: Using Computational Text Analysis to Measure Legal Importance

I used some of the computational text analysis tools outlined in the previous two articles in my latest article, *Measuring Legal Importance: From Case Citations to Linguistic Shifts*, 23 Nw. J. Tech. & Intell. Prop. 551 (2026). In this article, I developed a novel methodology for measuring judicial impact that goes beyond traditional citation counts. I used computational text analysis techniques such as topic modeling and Hellinger similarity distance to analyze linguistic patterns in hundreds of patents before and after a key Federal Circuit decision. My analysis revealed significant linguistic shifts in these patents, demonstrating that this under-cited decision was actually highly influential in shaping how practitioners draft claims. I argued that this approach allows legal scholars to identify impactful judicial decisions that traditional proxies of importance miss, broadening our understanding of how courts shape legal practice and society beyond the litigation context.

C. Current and Future Research

Which Cases Do Language Models Already Know?

In a coauthored project, *Which Cases Do Language Models Already Know?*, my coauthors and I examine whether language models used in legal research are reasoning from supplied materials or recalling previously encountered cases. We develop a 0–100 recognition score by giving a model only a case’s caption, court, and year and asking it to provide verifiable details it was not given, including the reporter citation, authoring judge, and disposition. Across six commercial models, recognition is concentrated among prominent and heavily cited decisions and is generally greater in newer frontier models. We also show that recognition can distort downstream empirical results: revealing the identity of a recognized case can increase appellate-outcome prediction accuracy by as much as thirty-six percentage points, while having no effect on unrecognized cases. The paper concludes with a practical protocol—screen, stratify, and falsify—for researchers using decided cases to evaluate legal AI or generate empirical measurements.